

дз на 21.10

P23) ✓3

- 1) efficiency, safety *scanning*
- 2) concept *measurement*
- 3) attach, sections
- 4) consists of
- 5) integrated, twin
- 6) integration

✓4

1. True
2. False
3. False *no roots*
4. True
5. False *no roots*
6. True

✓5

- 1) lead time
- 2) acquisition
- 3) assembly line
- 4) logistics
- 5) ergonomics
- 6) integration
- 7) digital twin

✓6

- 1) lead time
- 2) logistics
- 3) ergonomics
- 4) integration
- 5) digital

✓7

обсуждали в классе

P24)

✓2

- | | |
|----------------------|------------------------------------|
| • boron oxides ✓ | • nanosized zirconia |
| • carbon nanotubes ✓ | • polymers, metals, and ceramics ✓ |
| • epoxy/clay | • silicon oxides ✓ |
| • nanocomposites | • silver nanoparticles |
| • fullerenes | • steel |
| • graphene ✓ | • titanium-nickel alloys ✓ |
| • magnesium alloys ✓ | |

✓3

- 1) C
- 2) a
- 3) c
- 4) b
- 5) c
- 6) c
- 7) b

✓4

- 1) T
- 2) F
- 3) F
- 4) T
- 5) T
- 6) T

P25)

✓5

- 1) Stronger
- 2) Lighter
- 3) Durable
- 4) Mechanical
- 5) High-performance
- 6) Corrosion-resistant
- 7) Wear-resistant
- 8) Ecological

✓6

- 1) Carbon-fiber-components
- 2) titanium alloy
- 3) reinforced polymers
- 4) Systems in aircraft
- 5) coatings
- 6) materials
- 7) cryogenic pumps
- 8) innovations

✓7

Key applications
 Lightweight & High-Strength materials
 Enhanced Protective Coatings
 Improved Thermal Management
 Additive Manufacturing Capabilities
 Advanced Sensors and Electronics

Main challenges

Cost considerations

Manufacturing Challenges

Regulatory Barriers

Environmental and Health Impacts

data acquisition system - система сбора данных

✓7 (03)

industry: *+* construction, *+* design, *+* maintenance

the home: *+* security, *+* maintenance, *-* pests

business: *???* working with customers